MIL-P-60574B (AR)
AMENDMENT 2
1 August 1985
SUPERSEDING
AMENDMENT 1
17 March 1972

MILITARY SPECIFICATION

PROJECTILE, HEP-T, 105MM, M393A2
METAL PARTS

This Amendment forms a part of Military Specification MIL-P-60574B (MU) dated 1 November 1971 and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and agencies of the Department of Defense.

PAGE 2

* 2.1, Drawings, add: 8648732 - Eddy current Inspection System.

9280265 - Envelope Drawing, Ultrasonic Inspection Equipment for Projectile, 105MM, HEP-T, M393A2

9323896 - Envelope drawing, Ultrasonic Inspection Equipment for Projectile, 105MM, HEP-T, M393A2

PAGE 4

3.6, Delete "a. Permanent deformation greater than .010 inches on either the body diameter or bourrelet diameter of the projectile."

PAGE 8

* 4.3.2.2, Delete in its entirety and substitute:

"4.3.2.2 Plug, Base

Drawing No. 9335986
Next Higher Assembly 10424305

FSC 1315

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Category	Defect	AQL	Inspection Method
Critical			
1	Metal defective	200%	Visual (4X magnif)
2	Eddy current inspection		Test
3	Eddy current acceptance stamp missing		Visual
Major			
101	Runout of pitch diameter of large internal thread with pitch diameter of external thread, including front face of shoulder and bottom of large internal thread cavity	0.40%	Gage
102	Runout of large outside diameter with pitch diameter of external thread, including front face of shoulder	0.40%	Gage
103	Runout of pitch diameter of small internal thread with pitch diameter of external thread including front face of shoulder and rear face	0.40%	Gage
104	Runout of front face of shoulder with pitch diameter of external thread	0.40%	Gage
105	Presence and location of wrench holes	0.40%	Gage
106	Diameter of wrench holes	0.40%	Gage
107	Pitch diameter of external thread, min. at three equi-distant locations along length	100%	Gage
108	Major diameter of external thread, min.	0.40%	Gage
109	Radius between front face of shoulder and large external thread	0.40%	Gage

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Major (Continued)

110	Pitch diameter of large internal thread	0.40%	Gage
111	Minor diameter of large internal thread, max.	0.40%	Gage
112	Pitch diameter of small internal thread, min.	0.40%	Gage
113	Minor diameter of small internal thread, max.	0.40%	Gage
114	Length of effective small internal thread	0.40%	Gage
115	Length of ineffective external thread, max.	0.40%	Gage
116	Length of effective large internal thread, min.	0.40%	Gage
117	Depth from front of shoulder to bottom of large cavity	0.40%	Gage
118	Depth of tracer cavity, max.	0.40%	Gage
119	Depth of large internal cavity, max.	0.40%	Gage
120	Diameter of large internal cavity	0.40%	Gage
121	Thread(s) damaged or not full	0.40%	Visual
122	Surface finish of front face of shoulder improper	0.40%	Test
123	Corner chamfer, max.	0.40%	Gage
124	Depth of wrench holes	0.40%	Gage
125	Base plug tensile strength		Test
126	Protective coating corrosion test (salt spray)		Test

Minor

201	Diameter of shoulder	0.65%	Gage
202	Diameter at rear of taper	0.65%	Gage
203	Length from front face to shoulder	0.65%	Gage
204	Length, total	0.65%	Gage
205	Thickness of shoulder	0.65%	Gage
206	Radii or chamfers missing or incorrect	0.65%	Visual/Gage
207	Surface finish improper excluding front face of shoulder	0.65%	Test

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208	Protective coating inadequate or damaged	0.65%	Visual
209	Burr	0.65%	Visual/ Manual
210	Foreign matter	0.65%	Visual
211	Evidence of poor workmanship	0.65%	Visual".

PAGE 9

* 4.3.2.3:

In heading delete: "Part No." and substitute "Drawing No.".

Under Critical - Method of Inspection delete: "Visual (Probe light)" and substitute "Ultrasonic".

PAGE 10

* 4.3.2.4:

Under Critical - Method of Inspection delete" "Visual (Probe light)" and subsitute "Ultrasonic".

Delete Major's 101 through 109 in their entirety and substitute:

"101.	Runout of cavity forward of base plug thread with bourrelets	Gage
102.	Runout of forward rotating band with bourrelets	Gage
103.	Runout or rear rotating band with bourrelets	Gage
104.	Runout of pitch diameter of base plug thread with bourreletts including rear face	Gage
105.	True position of large counterbore with pitch diameter of thread	Gage
106.	True position of small counterbore with large counterbore	Gage
107.	Flatness of rear face	Gage
108.	Perpendicularity of rear face with pitch diameter of thread	Gage"

PAGE 12

4.3.2.5:

Add to Major characteristics:

108. Paint on rear face Visual"

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Delete: Minor characteristic 204 in its entirety.

PAGE 15

Add the following paragraph:

"4.3.3.1.2.6 Eddy current inspection. Each finish machined base plug shall be inspected by the eddy current method using standards prescribed by drawing 8648732. Failure of the base plug to meet the standard will be cause for rejection of the base plug."

PAGE 17

* Add the following paragraph:

"4.3.3.6 Ultrasonic inspection. Each projectile body 10524307 and projectile body assembly 10524306 shall be ultrasonically inspected. Failure of a body to meet the standard will be cause for rejection of the body."

PAGE 20

* Add the following paragraph:

"4.4.2.a Eddy current inspection. The procedure for performing the Eddy current test will be as prescribed by drawing 8648732. Eddy current standards shall be used at a minimum of every 4 hours. The standards shall be used at the start of each day's production and at intervals of not over 4 hours of continuous operation. The contractor shall also prepare a detailed written standard operating procedure to include the testing and calibration procedure which must be reviewed and approved by the responsible technical agency."

PAGE 21

* Add the following paragraph:

"4.4.7 Ultrasonic testing. Ultrasonic testing of the Body 10524307 and Body Assy 10524306 will be conducted in accordance with Envelope Drawings 9280265 and 9323986 respectively."

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5.2.2, Insert after the word packed: "Level C".

5.3.2: Delete "in accordance with paragraph 3.4" and substitute "Level C in accordance with paragraph 3.7".

NOTE: The margins of this amendment are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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